Assignment 1: Linear Algebra

Jack Gindi

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Problems are from Boyd & Vandenberghe, *Introduction to Applied Linear Algebra*. A freely available PDF is online — search for "VMLS Boyd Vandenberghe." Starred problems (*) are required. Others are optional but encouraged.

1 Vectors

- 1.3: Overloading*
- 1.4: Periodic energy usage*
- 1.5: Interpreting sparsity*
- 1.6: Vector of differences*
- 1.11: Word count and word histogram vectors*

2 Norms and Distances

- 3.4: Norm identities*
- 3.11: Neighboring electronic health records*

3 Matrices

- 6.1: Matrix and vector notation*
- 6.2: Matrix notation*
- 6.6: Matrix-vector multiplication*
- 7.3: Trimming a vector (Optional)

4 Matrix-Matrix Multiplication

- 10.2: ones matrix*
- 10.3: Matrix sizes*
- 10.17: Patients and symptoms*

5 Linear Equations

- 8.6: Linear functions*
- 8.9: Required nutrients (Optional)

6 Programming Assignment

See the `hw/` folder for the coding portion of this module. Run `uv run python score.py` to check your autograded score.